

	ITER, SIKSHA 'O' ANUSANDHAN Deemed to be University		Assignment-I
Programme	B.Tech.	Academic Year	2024-25
Department	Mathematics	Semester	2nd
Subject Code	MTH 2101	Marks	10
Subject Name	CALCULUS B		

Sl.No.	Questions	Taxonomy
Q.1	Show that the equation represents a sphere, and find its center and radius $x^2 + y^2 + z^2 + 8x - 6y + 2z + 17 = 0$.	L1
Q.2	Find an equation of a sphere if one of its diameters has endpoint (2,1,4) and (4,3,10).	L1
Q.3	What is the angle between the vector $8i + 6j$ and the positive direction of the x-axis.	L1
Q.4	State whether the given vectors $a = \langle 4, 6 \rangle$, $b = \langle -3, 2 \rangle$ are orthogonal, parallel, or neither	L1
Q.5	If a vector has direction angles $\alpha = \pi/4$ and $\beta = \pi/3$, find the third direction angle γ .	L1
Q.6	Use the scalar triple product to verify that the vectors $u = i + 5j - 2k$, $v = 3i - j$, and $w = 5i + 9j - 4k$ are coplanar.	L3
Q.7	Find symmetric equation for line of intersection of the planes $z = 2x - y - 5$, $z = 4x + 3y - 5$.	L1
Q.8	Find the equation of the normal plane and osculating plane of the curve $x = 2 \sin 3t$, $y = t$, $z = 2 \cos 3t$ at $(0, \pi, -2)$.	L1

Q.9	Find the curvature of $r(t) = \langle t, t^2, t^3 \rangle$ at the point (1, 1, 1)	L1
Q.10	Find equations of the normal plane and osculating plane of the curve $x = 2 \sin 3t, y = t, z = 2 \cos 3t$ at $(0, \pi, -2)$.	L1
Q.11	Find the tangential and normal components of the acceleration vector $r(t) = e^t i + \sqrt{2}tj + e^{-1}k$. GATE - 2022	L1
Q.12	Find the vector that is perpendicular to the vector $(i+j+k)$ and $(i+2j+3k)$ GATE - 2021	L1